C-Programming

Assignment for attendance on 24.04.2026

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Questions

1. Write a C program to find the maximum of three numbers without using logical operators.

Program

```c
#include <stdio.h>

int main() {
    int a, b, c, max;

    printf("Enter three integers: ");
    scanf("%d %d %d", &a, &b, &c);

    max = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);

    printf("The maximum number is: %d\n", max);

    return 0;
}
```

Output

minmax.c > main

```c
int main() {
    int a, b, c, max;

    printf("Enter three integers: ");
    scanf("%d %d %d", &a, &b, &c);

    max = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);

    printf("The maximum number is: %d\n", max);

    return 0;
}
```

Problems 3 Output Debug Console Terminal Ports

```
PS C:\Users\admin\C-Programming> .\minmax.exe
Enter three integers: 5 8 9
The maximum number is: 9
PS C:\Users\admin\C-Programming>
```

2. Write a C program to check whether the given year is leap year or not by adding century leap year or non-century leap year in the output (Eg: 2000 is a century leap year, 2024 is a non-century leap year)

Program

```c
#include <stdio.h>

int main() {
  int year;

  printf("Enter a year: ");
  scanf("%d", &year);

  if (year % 400 == 0) {
```

```
    printf("%d is a century leap year.\n", year);
  } else if (year % 100 == 0) {
    printf("%d is a century year, but NOT a leap year.\n", year);
  } else if (year % 4 == 0) {
    printf("%d is a non-century leap year.\n", year);
  } else {
    printf("%d is NOT a leap year.\n", year);
  }

  return 0;
}
```

Output

```
cen.c > main
#include <stdio.h>

int main() {
  int year;

  printf("Enter a year: ");
  scanf("%d", &year);

  if (year % 400 == 0) {
    printf("%d is a century leap year.\n", year);
  } else if (year % 100 == 0) {
    printf("%d is a century year, but NOT a leap year.\n", year);
  } else if (year % 4 == 0) {
    printf("%d is a non-century leap year.\n", year);
  } else {
    printf("%d is NOT a leap year.\n", year);
  }

  return 0;
}
```

```
Problems 15   Output   Debug Console   Terminal   Ports                powershell

PS C:\Users\admin\C-Programming> & "C:\Program Files (x86)\Embarcadero\Dev-Cpp\TDM-GCC-64\bin\gcc.exe" cen.c -o cen.exe
PS C:\Users\admin\C-Programming> .\cen.exe
Enter a year: 2006
2006 is NOT a leap year.
PS C:\Users\admin\C-Programming>
```

3. Write a C program to find whether the entered character is alphabet / digit / special character. If the entered character is an alphabet then say it is vowel or consonant without using built in functions.

Program

```
#include <stdio.h>

int main() {
  char ch;

  printf("Enter a character: ");
```

```c
  scanf(" %c", &ch);

  if (ch >= '0' && ch <= '9') {
    printf("'%c' is a digit.\n", ch);
  } else if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
        ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
      printf("'%c' is an alphabet and a vowel.\n", ch);
    } else {
      printf("'%c' is an alphabet and a consonant.\n", ch);
    }
  } else {
    printf("'%c' is a special character.\n", ch);
  }

  return 0;
}
```

Output

```
c char_id.c > main
#include <stdio.h>

int main() {
  char ch;

  printf("Enter a character: ");
  scanf(" %c", &ch);

  if (ch >= '0' && ch <= '9') {
    printf("'%c' is a digit.\n", ch);
  } else if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
        ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
      printf("'%c' is an alphabet and a vowel.\n", ch);
    } else {
      printf("'%c' is an alphabet and a consonant.\n", ch);
    }
  } else {
    printf("'%c' is a special character.\n", ch);
  }

  return 0;
}

Problems 15   Output   Debug Console   Terminal   Ports        powershell

PS C:\Users\admin\C-Programming> & "C:\Program Files (x86)\Embarcadero\Dev-Cpp\TDM-GCC-64\bin\gcc.exe" char_id.c -o char_id.exe
PS C:\Users\admin\C-Programming> .\char_id.exe
Enter a character: Gokul
'G' is an alphabet and a consonant.
PS C:\Users\admin\C-Programming> .\char_id.exe
Enter a character: I
'I' is an alphabet and a vowel.
PS C:\Users\admin\C-Programming>
```

4. Write a C program for simple ATM simulation with operations Check Balance, Deposit, Withdraw, Exit using switch and update balance accordingly.

Program

```c
#include <stdio.h>

int main() {
  int choice;
  float balance = 5000.0, amount;

  printf("1.Balance  2.Deposit  3.Withdraw\nChoice: ");
  scanf("%d", &choice);

  switch (choice) {
  case 1:
    printf("Balance: %.2f\n", balance);
    break;
  case 2:
    printf("Amount to deposit: ");
    scanf("%f", &amount);
    printf("New Balance: %.2f\n", balance + amount);
    break;
  case 3:
    printf("Amount to withdraw: ");
    scanf("%f", &amount);
    // Using a ternary operator to keep it to one line
    (amount <= balance) ? printf("New Balance: %.2f\n", balance - amount)
                        : printf("Insufficient funds.\n");
    break;
  default:
    printf("Invalid choice.\n");
  }
  return 0;
}
```

Output

atm.c > main

```c
#include <stdio.h>

int main() {
  int choice;
  float balance = 5000.0, amount;

  printf("1.Balance  2.Deposit  3.Withdraw\nChoice: ");
  scanf("%d", &choice);

  switch (choice) {
  case 1:
    printf("Balance: %.2f\n", balance);
    break;
  case 2:
    printf("Amount to deposit: ");
    scanf("%f", &amount);
    printf("New Balance: %.2f\n", balance + amount);
    break;
  case 3:
    printf("Amount to withdraw: ");
    scanf("%f", &amount);
    // Using a ternary operator to keep it to one line
    (amount <= balance) ? printf("New Balance: %.2f\n", balance - amount)
                        : printf("Insufficient funds.\n");
    break;
  default:
    printf("Invalid choice.\n");
  }
  return 0;
}
```

Problems 15 Output Debug Console Terminal Ports

```
PS C:\Users\admin\C-Programming> .\atm.exe
1.Balance  2.Deposit  3.Withdraw
Choice: 2
Amount to deposit: 300
New Balance: 5300.00
PS C:\Users\admin\C-Programming> .\atm.exe
1.Balance  2.Deposit  3.Withdraw
Choice: 3
Amount to withdraw: 900
New Balance: 4100.00
PS C:\Users\admin\C-Programming>
```

5. Write a C program for menu driven calculator using switch statement.

Program

```c
#include <stdio.h>

int main() {
  int op;
  float a, b;

  printf("1.Add  2.Subtract  3.Multiply  4.Divide\nChoice: ");
  scanf("%d", &op);

  printf("Enter two numbers: ");
  scanf("%f %f", &a, &b);

  switch (op) {
  case 1:
    printf("Result: %.2f\n", a + b);
    break;
  case 2:
    printf("Result: %.2f\n", a - b);
    break;
  case 3:
    printf("Result: %.2f\n", a * b);
    break;
  case 4:
    (b != 0) ? printf("Result: %.2f\n", a / b)
             : printf("Error: Div by zero.\n");
    break;
  default:
    printf("Invalid choice.\n");
  }
  return 0;
}
```

Output

calc.c > main

```c
#include <stdio.h>

int main() {
  int op;
  float a, b;

  printf("1.Add  2.Subtract  3.Multiply  4.Divide\nChoice: ");
  scanf("%d", &op);

  printf("Enter two numbers: ");
  scanf("%f %f", &a, &b);

  switch (op) {
  case 1:
    printf("Result: %.2f\n", a + b);
    break;
  case 2:
    printf("Result: %.2f\n", a - b);
    break;
  case 3:
    printf("Result: %.2f\n", a * b);
    break;
  case 4:
    (b != 0) ? printf("Result: %.2f\n", a / b)
             : printf("Error: Div by zero.\n");
    break;
  default:
    printf("Invalid choice.\n");
  }
  return 0;
}
```

Problems 15 Output Debug Console Terminal Ports

```
PS C:\Users\admin\C-Programming> .\calc.exe
1.Add  2.Subtract  3.Multiply  4.Divide
Choice: 1
Enter two numbers: 2 7
Result: 9.00
PS C:\Users\admin\C-Programming> .\calc.exe
1.Add  2.Subtract  3.Multiply  4.Divide
Choice: 2
Enter two numbers: 27 08
Result: 19.00
PS C:\Users\admin\C-Programming>
```